GG-C-106A
October 12, 1977
SUPERSEDING
Fed. Spec. GG-C-106
August 11, 1961

FEDERAL SPECIFICATION

CALIPER, EYE, CASTROVIEJO

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of eye caliper of the Castroviejo design, suitable for use in measuring distances on the eye ball.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

Federal Specifications:

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6515

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifiations, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof; Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, Standards, and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E92 - Vickers Hardness of Metallic Materials.
ASTM E140 - Standard Hardness Conversion Tables for Metals.

3. REQUIREMENTS

3.1 General requirements. The eye caliper shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

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3.2 Material. The instrument in its entirety, except for the spring, screw, nut and stud shall be fabricated of materials specified in Table I and in accordance with the applicable requirements of GG-I-526. The spring shall be fabricated of AISI Type 300 series stainless steel hardened to provide sufficient spring action. The remainder of the instrument (screw, nut and stud) shall be fabricated of the appropriate AISI Type 400 series stainless steel.

3.2.1 Hardness. After proper stress relieving the hardness shall be as specified in Table I when tested as specified in 4.4.1.

TABLE I. Material and hardness requirements.

Component	Class	Type	Hardness
Scale	3	302, 303 or 304 1/	B75 to B100
	4	420	C40 to C50
Shanks (including points)	4	410, 420	C40 to C50

1/ As an alternate the chromium content shall be 17.00 to 20.00 percent.

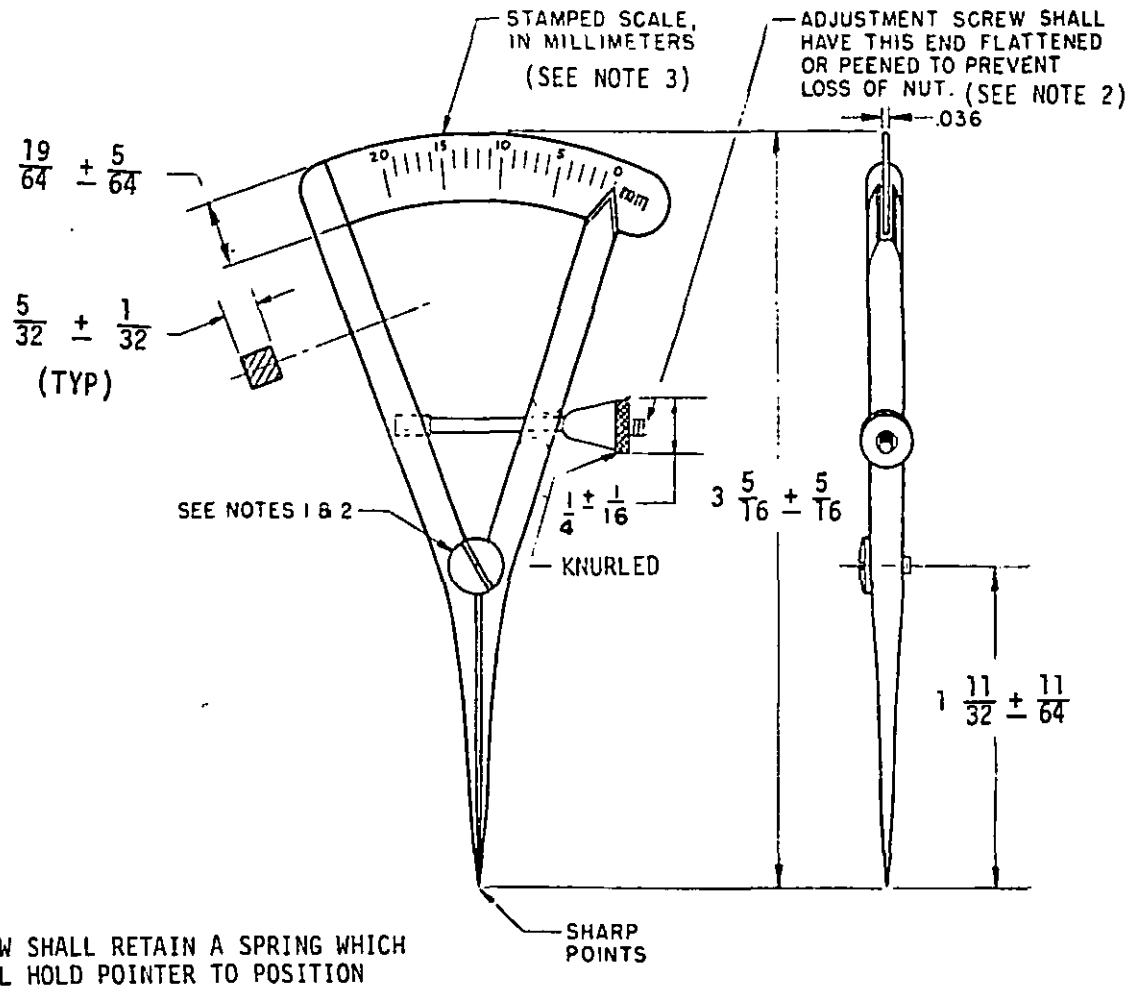
3.3 Style, design and dimensions. The caliper shall be of the Castroviejo design, suitable for measuring distances on the eyeball. The caliper shall be spring loaded and shall have sharp points. The scale shall be calibrated from 0 to 20 millimeters (mm) in 1 mm intervals. The 0, 5, 10, 15 and 20 mm graduation marks shall be identified. The graduation marks and numerals shall be metal stamped. The calipers shall be in accordance with figure 1.

3.4 Accuracy. The caliper shall be accurate to within ± 0.5 millimeters (mm) over the range of 1 mm to 20 mm when tested as specified in 4.4.2.

3.5 Finish. The scale shall have a satin finish and the remainder of the instrument shall have a mirror finish.

3.6 Identification markings. The caliper shall be marked with the name or registered trademark of the contractor. Calipers of other than U.S. manufacture shall also be marked stainless.

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NOTES

1. SCREW SHALL RETAIN A SPRING WHICH SHALL HOLD POINTER TO POSITION SET BY ADJUSTMENT SCREW.
2. SCREW SHALL NOT BIND OR LOOSEN DURING USE. SCREW LOCK SHALL BE TIGHT ENOUGH TO PREVENT WOBBLING OF POINTS.
3. AS AN OPTION THE SCALE MAY APPEAR ON BOTH SIDES.

ALL DIMENSIONS ARE IN INCHES. TOLERANCES ON DECIMALS ± .005

FIGURE 1. Eye caliper, Castroviejo.

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3.7 Workmanship. Workmanship shall be first class throughout. Instruments shall be free from defects which detract from their appearance or impair their serviceability, proper functioning and intended use. Instrument shall be furnished free of crevices, cracks, pits, burrs, scale and corrosion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for examination purposes shall be one caliper.

TABLE II. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table III. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one caliper.

TABLE III. Sampling for tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL
1 - Caliper	Hardness	3.2.1	4.4.1	S-2	1.0
	Dimensions	3.3	--------	S-2	2.5
	Boil Test 1/	GG-I-526	GG-I-526	S-2	1.0
	Copper sulfate	GG-I-526	GG-I-526	S-2	1.0
	Accuracy	3.4	4.4.2	S-2	2.5

1/ Components fabricated from type 400 series material only.

4.2.3 Material composition. A sufficient quantity of each lot of metal used in the manufacture of the caliper shall be selected for composition analysis in accordance with GG-I-526.

4.3 Examination. The caliper shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables II and IV. Examination, however, shall not be restricted to defects listed in table IV. Any caliper in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

TABLE IV. Classification of defects.

Categories	Defects
Major:	
101	Style and design not as specified.
102	Caliper not complete.
103	Caliper not free of sharp edges, rough areas, burrs, cracks, grind marks, or overlaps.
104	Tips not sharp.
105	Spring does not hold caliper at set position.
Minor:	
201	Caliper not free from foreign matter.
202	Finish not as specified.

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4.3.1 Examination*Workmanship shall be first class throughout. Instruments* performed to determinects which detract from their appearance or impair their requirements of secti functioning and intended use. Instrument shall be Sampling shall be in ices, cracks, pits, burrs, scale and corrosion. one container fully p containers offered foCE PROVISIONS be S-2 with an AQL of

for inspection. Unless otherwise specified in the

TABLE V. or is responsible for the performance of all inspection

Examination	ied herein. Except as otherwise specified in the contract,
	his own or any other facilities suitable for the
Containers	Notpection requirements specified *herein, unless disapproved*
	e Government reserves the right to perform any of the
Content	Notin the specification where such inspections are deemed
	pplies and services conform to prescribed requirements.
Marking	Omi
	se Inspection shall be in accordance with GG-I-526.
Material	Comcords of *examinations and tests performed by or for the*
	intained by the contractor and made available to the
Workmanship	Bulovernment's request, at any time, or from time to time,
	Cus of the contract and for a period of three years after
	es to which such records relate.

4.3.1.1 Examinati of quality. Certificates of quality, supplied by the *containers are requir*terials may be furnished in lieu of actual performance PPP-B-636 or PPP-B-67 contractor, provided lot identity has been maintained banding shall be in ad to the Government. The certificate shall include ctor, the contract number, the name of the manufacturer

4.4 Tests. Tests the Item Identification, the name of the component/ cation requirements. er, the lot size, the sample size, the date of testing, idual test results, and the specification requirements.

4.4.1 Hardness te or ASTM E92. Convers shall be in accordanc

ion. Sampling for examination shall be conducted in

4.4.2 Accuracy teD-105 and as indicated in table II. The unit of product scale. Read the calies shall be one caliper. the caliper points to requirements of 3.4.

LE II. Sampling for examination.

	Inspection level	AQL (percent defective)
	S-4	1.0
	S-4	2.5

5. PREPARATION FO

5.1 Packaging. P

5.1.1 Level A.

5.1.1.1 Cleaning dried in accordance w

5.1.1.2 Preservat

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5.1.1.3 Unit package. Each caliper shall be wrapped in neutral tissue or other suitable non-corrosive material. The caliper shall be further packaged in a box conforming to PPP-B-566 or PPP-B-676. The box shall contain sufficient cushioning material or designed with suitable interior fittings to prevent damage to the points of the calipers. Closure shall be as specified in the box specification.

5.1.1.4 Intermediate package. Twelve (12) calipers shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Calipers shall be clean and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two hundred eighty-eight (288) calipers shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 3 style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Two hundred eighty-eight (288) calipers shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged calipers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

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5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The eye caliper covered by this specification is intended for use in measuring distances on the eye ball.

6.2 Ordering data. Purchasers should exercise the perferred options offered herein, and include the following information in the procurement documents.

(a) Number, title and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(c) Marking required (see 5.3).

6.3 This specification is intended to cover the following item:

National Stock Number	Item Identification
6515-00-299-9497	Caliper, Eye, Castroviejo

6.4 Supersession data. This specification includes the requirements of Military Specification MIL-C-37044 dated 9 April 1974 for Caliper, Eye, Castroviejo.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Civil Agency Coordinating Activities:
GSA-FSS
HEW-FDA

Project No. 6515-2092

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MATERIAL PROCURED UN

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